

Shubham Sarkar

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Education

Jadavpur University Bachelor of Technology	Nov 2023 – Dec 2027 CGPA: 7.5
Hariyana Vidya Mandir Higher Secondary	Apr 2020 – Apr 2022 Percentage: 90%

Experience

Alignerr Machine Learning Engineer	Jan 2026 – Present Remote
- Enhanced LLM evaluation precision by 15% through a comprehensive review of a rubric-based scoring framework across 6 reasoning categories. - Analyzed over 50 audio files for integration into ASR pipelines, improving data ingestion processes. - Evaluated AI agent responses to identify failure points such as inference memory and self-coherence, leading to a 20% improvement in model accuracy.	
Jadavpur University CMATER Lab Deep Learning Research Assistant	May 2025 – September 2025 Kolkata, India
- Designed and implemented a self-attention mechanism (scaled dot-product) within a pre-trained VGG16, significantly enhancing feature extraction for lung cancer detection from CT scans. - Developed a hybrid deep learning architecture achieving 99.54% peak accuracy with only 76k trainable parameters and 0.0256 GFLOPs, facilitating edge-device deployment. - Engineered feature fusion through concatenation and element-wise multiplication of original and attention-modulated maps for refined, context-aware representations.	

Projects

TailorCV.ai Python, FastAPI, LLM, AI agents, Amazon Web Services GitHub ↗ Live Demo ↗	2 January 2026
- Developed an AI web application that optimizes resumes to job descriptions using LLMs and NLP pipelines, improving resume relevance by up to 80%. - Designed a Python and FastAPI backend integrated with HTML, CSS, and JavaScript for the frontend, dockerized the application, and deployed it on AWS ECS. - Achieved over 50 users within the first week of launch, demonstrating strong early adoption and real-world impact.	
YouTube Sentiment Analysis TensorFlow, NLP, AWS EC2, Scikit-learn	2 October 2025
- Created an end-to-end YouTube sentiment analysis pipeline processing over 10,000 user comments, enhancing sentiment classification performance through NLP preprocessing techniques. - Tracked multiple model experiments using MLflow and DVC, enabling reproducible training and systematic comparison of models built with scikit-learn and NLP libraries. - Deployed the pipeline on AWS using Docker and exposed predictions via Flask REST APIs, facilitating scalable and reproducible inference.	
Smart Product Pricing Keras, Hugging Face Transformers, ResNet50, OpenCV Kaggle ↗	2 October 2025
- Developed an NLP and CV pipeline to analyze 150,000 image and text data using transformer-based text encoders and CNN-based image embeddings, integrating them through a fusion neural network for price prediction. - Implemented data preprocessing techniques, including text cleaning, tokenization, and streaming image feature extraction with ResNet and CLIP representations to manage large datasets. - Built and fine-tuned models using TensorFlow and scikit-learn, achieving a rank of 142 out of 50,000 participants.	

Technical Skills

Languages: SQL, Python, C, JavaScript, HTML, CSS
AI/ML: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, LLM Fine Tuning, RAG, LLM Frameworks/Libraries: TensorFlow, PyTorch, scikit-learn, Hugging Face, Pandas, NumPy, Langchain, Flask, FastAPI, Streamlit, ETL Pipelines
Databases: PostgreSQL, Vector Databases, Chroma
Tools & Platforms: MLflow, DVC
Cloud & DevOps: Amazon Web Services, AWS, Docker, CI/CD, Linux
Other Technical Skills: Data Structures and Algorithms, OpenAI, tool-using LLM systems

Leadership / Extracurricular

Core Member at Entrepreneurship Cell, Jadavpur University

May 2024 – Present

- Organized national-level events such as E-Summit 2025 and Hult Prize 2025, attracting over 5,000 registrations and 1,000+ attendees.
- Contributed to the establishment of an Incubation Center at Jadavpur University under the Institution’s Innovation Council (IIC).